

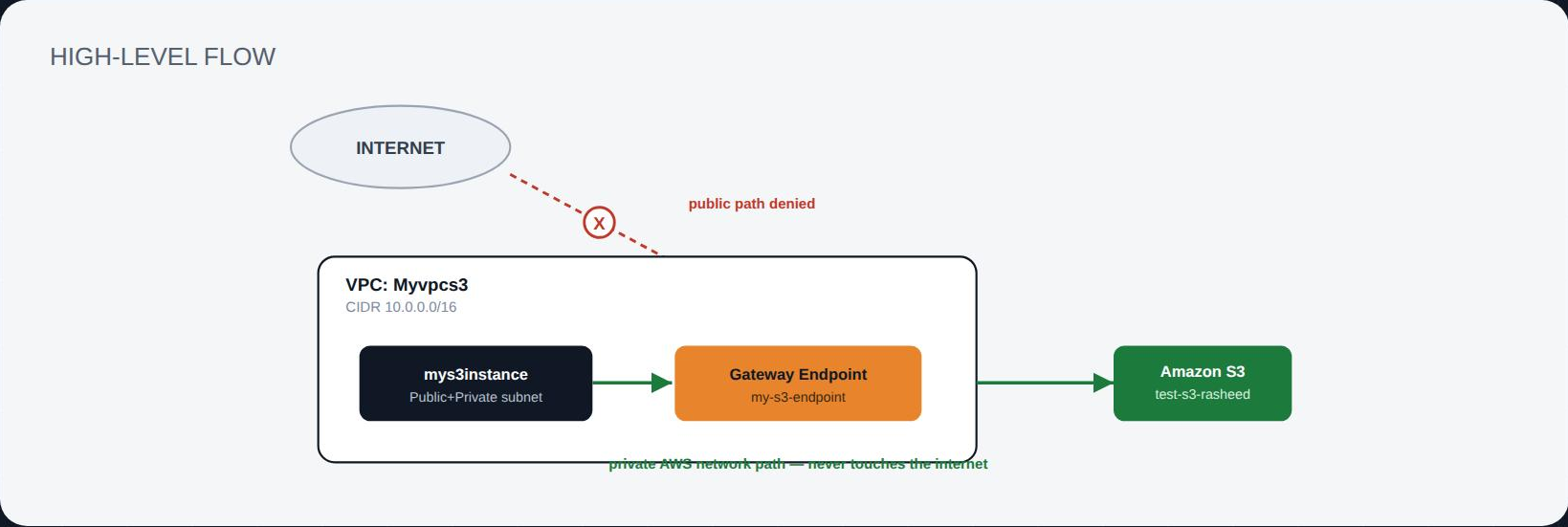


linkedin.com/in/
mohammed-abdul-
rasheed

Securing S3 Access with a VPC Gateway Endpoint

Moving S3 access off the public internet path and onto AWS's private backbone — then proving it with a real bucket policy that denies every request unless it arrives through the endpoint, and a live terminal test that fails before the fix and succeeds after it.

- 1 VPC
- 1 Gateway Endpoint
- 1 Route Table Entry
- 1 Bucket Policy
- 1 EC2 Instance
- Verified via CLI



Project Overview

This project shows why traffic between an EC2 instance and an S3 bucket doesn't have to leave AWS's network at all. Starting point: a VPC with a working Internet Gateway, where an EC2 instance could already reach S3 **over the public internet** using its IAM credentials. The goal was to close that public path entirely and force all S3 traffic through a **VPC Gateway Endpoint** instead — then prove, with a real bucket policy and a live terminal test, that access only works when it arrives through the endpoint.

Component	Name	Configuration
VPC	Myvpcs3	10.0.0.0/16
Subnet	my-s3-subnet	10.0.1.0/24 (us-east-1a)
Route Table	my-s3-route-table	0.0.0.0/0 → IGW, 10.0.0.0/16 → local, S3 prefix list → Endpoint
VPC Endpoint	my-s3-endpoint	Gateway type, service com.amazonaws.us-east-1.s3
EC2 Instance	mys3instance	Public 44.197.186.97 / Private 10.0.1.160
S3 Bucket	test-s3-rasheed	Bucket policy denies access outside the endpoint
IAM User	tests3	Configured on the instance via aws configure

Why this matters: Without an endpoint, traffic from a private EC2 instance to S3 either needs a public IP + Internet Gateway/NAT, or it silently routes out over the public AWS edge. A Gateway Endpoint keeps that traffic on Amazon's internal network, removes the need for internet egress just to reach S3, and lets a bucket policy enforce "only reachable through this endpoint" as a hard security boundary — not just an IAM permission.

- Free — no hourly charge for Gateway Endpoints
- No NAT Gateway required
- Enforced at the bucket policy layer

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Baseline — S3 Reachable Over the Internet

VPC · EC2 · Internet Gateway already in place

STEP 01

Confirm the VPC and instance

The starting point was a working VPC (**Myvpcs3**, `10.0.0.0/16`) with an Internet Gateway attached, and an EC2 instance (**mys3instance**) launched with a public IP `44.197.186.97` and private IP `10.0.1.160` , sitting in **my-s3-subnet**.

Your VPCs

VPCs

VPC encryption controls

Your VPCs (1)

Info

Last updated 6 minutes ago

Actions

Create VPC

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	Myvpcs3	vpc-044a85eb804364f08	Available	-	-	Off	10.0.0.0/16	-

Available

10.0.0.0/16

STEP 02

Launch the instance inside the S3 subnet

mys3instance was launched as a t3.micro in **my-s3-subnet**, picking up private IP `10.0.1.160` and public IP `44.197.186.97` — reachable over the internet and, at this point, still routing to S3 the same way: out through the Internet Gateway.

Instances (1/1)

Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☒	mys3instance	i-0efc2d0c680638393	Running	t3.micro	3/3 checks passed	us-east-1a	-	44.197.186.97	-	-

i-0efc2d0c680638393 (mys3instance)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

VPC ID

[vpc-044a85eb804364f08](#) (Myvpcs3)

Subnet ID

[subnet-083ab1313decc12ad](#) (my s3-subnet)

Availability zone

[us-east-1a](#)

Availability zone ID

[use1-az1](#)

Outpost ID

-

IP addresses

Info

Public IPv4 address

[44.197.186.97](#) | open address

Private IPv4 addresses

[10.0.1.160](#)

IPv6 addresses

-

Secondary private IPv4 addresses

-

Carrier IP addresses (ephemeral)

-

Running

3/3 checks passed

2

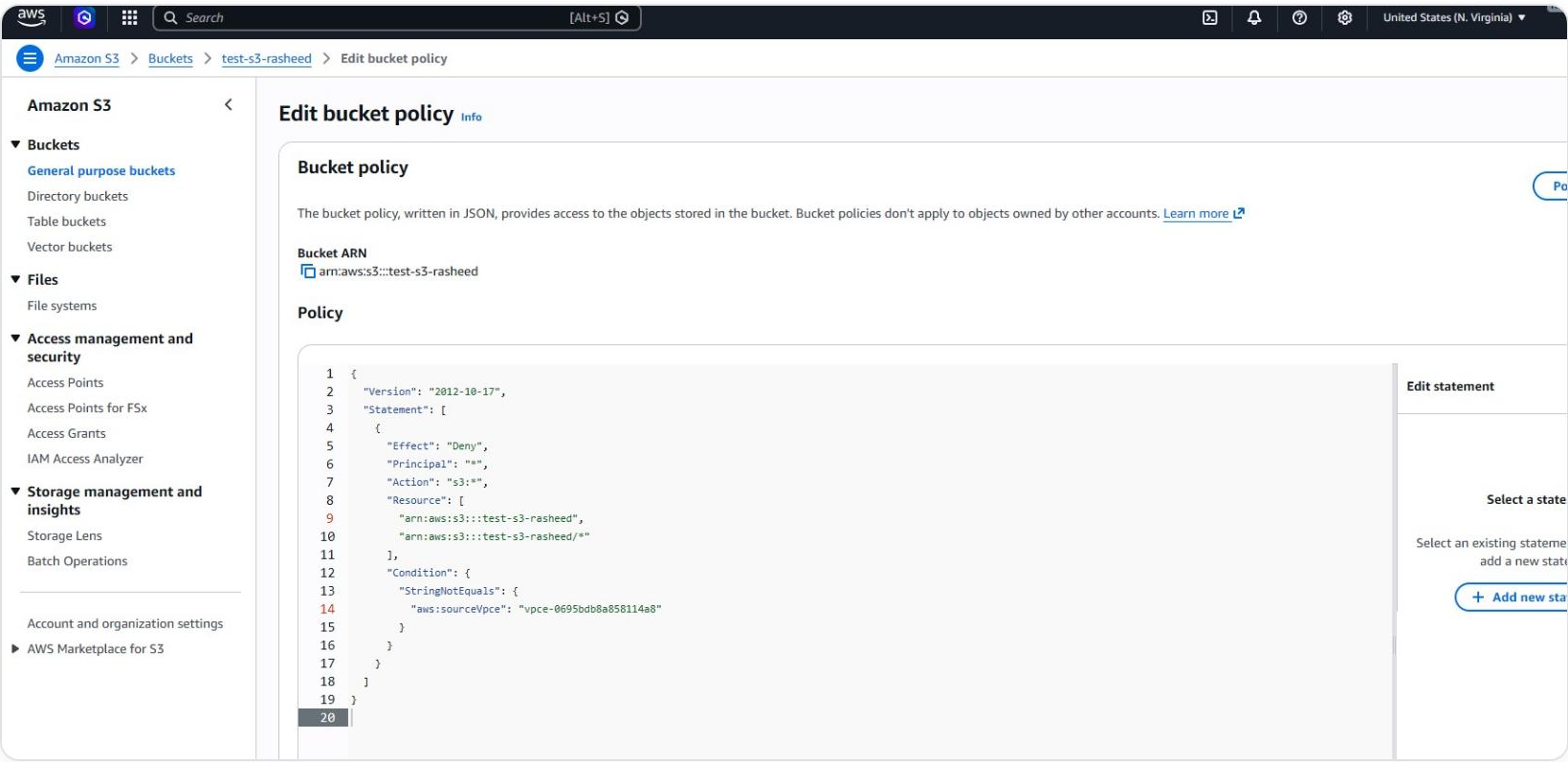
Lock the Bucket to the Endpoint

Bucket Policy · Deny outside the VPC Endpoint

STEP 03

Attach a deny-by-default bucket policy

A bucket policy was attached to `test-s3-rasheed` that **denies every S3 action** on the bucket and its objects unless the request's `aws:sourceVpce` matches the endpoint ID. At this stage the policy was written referencing the endpoint before it was fully wired up, so any access — including from inside the VPC — was rejected.



STEP 04

Test from the instance — access denied

Back on `mys3instance`, running `aws s3 ls s3://test-s3-rasheed` after the policy was attached returned **AccessDenied** — an explicit deny in a resource-based policy — because traffic was still leaving through the Internet Gateway, not the (not-yet-connected) endpoint. This confirms the policy is doing real enforcement, not just sitting there unused.



AccessDenied

Explicit deny — resource policy

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Create the Gateway Endpoint & Wire the Route

VPC Endpoint · Route Table Association

STEP 05

Create the S3 Gateway Endpoint

A VPC Endpoint named **my-s3-endpoint** was created for service `com.amazonaws.us-east-1.s3`, type **Gateway**, attached to **Myvpcs3**. Unlike an Interface Endpoint, a Gateway Endpoint doesn't get an IP — it works by inserting a special route into whichever route tables it's associated with.

Endpoints (1/1) Info

Find endpoints by attribute or tag

Name	VPC endpoint ID	Endpoint type	Status	Service name
my-s3-endpoint	vpce-0695bdb8a858114a8	Gateway	Available	com.amazonaws.us-east-1.s3

vpce-0695bdb8a858114a8 / my-s3-endpoint

Details

Endpoint ID

vpce-0695bdb8a858114a8

VPC ID

vpce-044a85eb804364f08 (Myvpcs3)

DNS record IP type

service-defined

Private DNS specified domains

-

Status

Available

Status message

-

IP address type

ipv4

Creation time

Wednesday, July 22, 2026 at 08:58:40 GMT+3

Service name

com.amazonaws.us-east-1.s3

Service region

us-east-1

Endpoint type

Gateway

Private DNS names enabled

No

Private DNS preference

-

Available Gateway type

STEP 06

Associate the endpoint with the route table

Associating **my-s3-endpoint** with **my-s3-route-table** added a new route: destination is the S3 **prefix list** (`pl-63a5400a`), target is the endpoint itself. Now any traffic to S3's IP ranges from this subnet is routed to the endpoint instead of out through the Internet Gateway — the private path now exists alongside the public one.

Route tables (1/2) Info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
my-s3-route-table	rtb-0d1c38d01f5e3e358	subnet-083ab1313decc1...	-	No	vpce-044a85eb804364f08 Myv...
-	rtb-004f36472a0d0a5fd	-	-	Yes	vpce-044a85eb804364f08 Myv...

rtb-0d1c38d01f5e3e358 / my-s3-route-table

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (3)

Filter routes

Destination	Target	Status	Propagated
pl-63a5400a	vpce-0695bdb8a858114a8	Active	No
0.0.0.0/0	igw-0e4dc758c9e3de757	Active	No
10.0.0.0/16	local	Active	No

pl-63a5400a → vpce Active

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Validate — Access Restored Through the Endpoint

CLI Test · Final Bucket Policy

STEP 07

Re-run the CLI test — it works

With the route in place, running `aws s3 ls s3://test-s3-rasheed` again from **mys3instance** now lists the bucket's objects successfully — traffic is leaving through the Gateway Endpoint, matching the `aws:sourceVpce` condition in the bucket policy, and the deny no longer applies.

aws

Search

[Alt+S]

Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

[REDACTED]

last login: Wed Jul 22 06:09:10 2026 from 18.206.107.27

ec2-user@ip-10-0-1-160 ~]\$ aws configure

AWS Access Key ID [*****TJQH]: AKIAYAMRPP3NAIQ5TJQH

AWS Secret Access Key [*****W8Kd]: tojqH51PIjCbCKintpg616gyp:WJDs97ATvDW8Kd

Default region name [us-east-1]: Default output format [None]:

ec2-user@ip-10-0-1-160 ~]\$ aws s3 ls

2026-07-21 05:56:27 test-s3-rasheed

ec2-user@ip-10-0-1-160 ~]\$ aws s3 ls s3://test-s3-rasheed

An error occurred (AccessDenied) when calling the ListObjectsV2 operation: User: arn:aws:iam::550597787354:user/tests3 is not authorized to perform: s3:ListBucket on resource: "arn:aws:s3:::test-s3-rasheed" due to an explicit deny in a resource-based policy

ec2-user@ip-10-0-1-160 ~]\$ ^C

ec2-user@ip-10-0-1-160 ~]\$ aws s3 ls s3://test-s3-rasheed

2026-07-21 05:56:51 455154 1.jpg

ec2-user@ip-10-0-1-160 ~]\$

i-0efc2d0c680638393 (mys3instance)

PublicIPs: 44.197.186.97 PrivateIPs: 10.0.1.160

- 1.jpg listed
- Access confirmed

STEP 08

Final bucket policy

The bucket's final state: a **Deny** statement on `s3:*` for the bucket and all objects in it, scoped by a `StringNotEquals` condition on `aws:sourceVpce`. Any request that doesn't arrive through **vpce-0695bdb8a858114a8** is rejected outright, regardless of IAM permissions.

Amazon S3 > Buckets > test-s3-rasheed

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Deny",
      "Principal": "*",
      "Action": "s3:*",
      "Resource": [
        "arn:aws:s3:::test-s3-rasheed",
        "arn:aws:s3:::test-s3-rasheed/*"
      ],
      "Condition": {
        "StringNotEquals": {
          "aws:sourceVpce": "vpce-0695bdb8a858114a8"
        }
      }
    }
  ]
}
```

Copy

Object Ownership

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Edit

Security note: The AWS Access Key ID and Secret Access Key visible in the original terminal screenshots have been redacted in this report before publishing.

Conclusion & Key Takeaways

Successfully closed off S3's public-internet path and proved that a private EC2 instance can only reach the bucket through a VPC Gateway Endpoint — enforced independently by routing and by the bucket policy itself, and validated with real, before-and-after CLI tests rather than just inspecting console screens.

Traffic Never Leaves AWS

Once the Gateway Endpoint route is in place, S3 traffic from the private subnet travels entirely over Amazon's internal network instead of the public internet.

Policy-Enforced, Not Just Routed

The bucket policy's `aws:sourceVpce` condition makes the endpoint mandatory — even valid IAM credentials fail if the request doesn't arrive through it.

No NAT Gateway Needed

A Gateway Endpoint is free and needs no NAT Gateway or extra public IPs just to let private resources reach S3.

Validated, Not Assumed

The design was proven end-to-end with a real CLI test: denied before the endpoint route existed, working after — not just inspected in the console.